# The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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## Abstract

We formalize the conceptual incompatibility between General Relativity (GR) and Quantum Field Theory (QFT) by identifying a fundamental structural asymmetry: standard quantum interactions represent local operator fields acting within a kinematic background, whereas gravitational dynamics dictate the background metric structure itself. We state this architectural divergence as the *Gravitational Baseline Principle*. We demonstrate that treating gravity as a conventional, quantizable gauge field introduces unavoidable mathematical contradictions regarding background independence and gauge covariance. To resolve this, we propose a minimal, background-independent mathematical framework in which both smooth Riemannian manifolds and quantum field algebras co-emerge from a discrete, relational quantum substrate governed by density-matrix dynamics. In this framework, the graviton is recovered not as an elementary gauge boson, but as a collective, low-energy excitation of the pre-geometric medium.

## 1 Introduction

Quantum Field Theory (QFT) describes matter and radiation via operator-valued distributions evolving on a fixed, smooth kinematic background [2]. General Relativity (GR), by contrast, denies the existence of any prior background container, asserting that spacetime geometry is a dynamical physical degree of freedom governed by the distribution of stress-energy [3]. This architectural mismatch underlies the century-long difficulty in formulating a consistent quantum theory of gravity [1].

Gravity is not a selective force acting upon a subset of fields; it is the universal geometric framework that defines causal and metrological relationships between all fields. We formalize this insight via the *Gravitational Baseline Principle* and outline the minimal mathematical constraints required for a pre-geometric quantum substrate to generate smooth continuous physical manifolds.

## 2 The Structural Asymmetry Problem

Einstein's field equations,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{1}$$

dictate that all physical fields couple universally to the metric tensor $g_{\mu\nu}$. In contrast, gauge interactions operate only on specific internal symmetry representations. Standard perturbative quantization techniques require a fixed background metric to define microcausality and construct a Fock space. If the metric is itself perturbed and quantized as $h_{\mu\nu}$, the framework violates diffeomorphism invariance at the foundational level.

## 3 The Gravitational Baseline Principle

Gravity is not a peer field to the gauge forces within a spacetime container, but the geometric framework that renders gauge field relationships definable. Because all physical systems couple non-linearly to this background, gravity serves as the absolute measurement baseline. The Planck scales,

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}, \tag{2}$$

mark the breakdown of the continuum description of geometry, exposing the pre-geometric substrate.

# 4 Formal Substrate Architecture

We define the pre-geometric substrate $Q$ as an abstract quantum system described by a state vector $|\Psi\rangle$ in a non-local, background-independent Hilbert space $H_Q$. The geometric connectivity of the emergent space is encoded in the entanglement structure of the state. Let $\rho_0 = |\Psi\rangle\langle\Psi|$ be the density matrix of the substrate. The mutual information

$$I(i,j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{3}$$

defines a symmetric relational connectivity matrix $A_{ij}$.

## 4.1 Emergent Continuum Limit

To reconstruct a smooth Riemannian spatial manifold from the relational matrix $A_{ij}$, we examine the action of a discrete structural operator mapping the subsystems:

$$\Delta f_i = \sum_j A_{ij}(f_i - f_j) \tag{4}$$

By mapping a smooth continuous test scalar field $\phi(x)$ to the discrete subsystem nodes via $f_i \equiv \phi(x_i)$ and evaluating a localized Taylor expansion, we obtain:

$$\Delta\phi = -\left(\sum_j A_{ij}\Delta x^\mu\right)\partial_\mu\phi - \frac{1}{2}\left(\sum_j A_{ij}\Delta x^\mu \Delta x^\nu\right)\partial_\mu\partial_\nu\phi \tag{5}$$

Statistical relaxation of the substrate minimizes non-isotropic configurations, enforcing the constraints $\sum_j A_{ij}\Delta x^\mu = 0$ and $\sum_j A_{ij}\Delta x^\mu \Delta x^\nu \to a^2\delta^{\mu\nu}$, where $\delta^{\mu\nu}$ is the emergent Riemannian metric tensor and $a$ is the relational scale length. Thus, the continuous spatial Laplace-Beltrami operator emerges cleanly as an expectation value: $\lim_{a\to 0}\frac{2}{a^2}\langle\Delta\rangle = \nabla^2$.

## 4.2 Dynamical Law of the Substrate

We introduce a minimal, background-independent dynamical law:

$$\frac{d\rho_0}{dt} = -i[H_Q, \rho_0]. \tag{6}$$

The Hamiltonian is constructed from the relational matrix:

$$H_Q = \sum_{i,j} K(A_{ij})\,\sigma_i \otimes \sigma_j, \tag{7}$$

where $K$ is a monotonic kernel of entanglement strength and $\sigma_i$ represents localized operator algebras acting on the modular subsystems. We note that because $H_Q$ depends implicitly on the state $\rho_0$ via the mutual information matrix $A_{ij}$, Equation (6) represents a self-consistent, non-linear state evolution equation of the generalized Schrödinger-Newton type. This structure ensures non-local microscopic dynamics, with locality emerging only when $A_{ij}$ becomes band-limited.

# 5 Theoretical Constraints

## 5.1 The Status of the Graviton

We consider small perturbations of the emergent metric around a flat background solution: $g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x)$, with $|h_{\mu\nu}| \ll 1$. The Pauli–Fierz action for a free, massless spin-2 field is:

$$S_{\mathrm{PF}}[h] = \int d^4x \left[-\frac{1}{2}\partial_\lambda h_{\mu\nu}\partial^\lambda h^{\mu\nu} + \partial_\mu h^{\mu\nu}\partial_\lambda h^\lambda_\nu - \partial_\mu h^{\mu\nu}\partial_\nu h + \cdots \right. \tag{8}$$

with $h = h^\mu_\mu$. In the present model, $h_{\mu\nu}$ arises from small variations of the relational matrix projected via smooth vector-valued coordinate projection functions $f^{(i)}_\mu(x)$ mapping the discrete nodes to the manifold:

$$h_{\mu\nu}(x) \sim \sum_{i,j} \delta A_{ij}\,\partial_\mu f^{(i)}(x)\,\partial_\nu f^{(j)}(x) \tag{9}$$

After projection onto the symmetric, transverse, traceless sector, the effective action reduces to the Pauli–Fierz form, proving the graviton is a collective low-energy excitation.

## 5.2 Thermodynamic Gravitational Dynamics

Einstein's equations emerge as a thermodynamic equation of state [4]. Fluctuations of the substrate generate higher-order curvature corrections:

$$S_{\mathrm{eff}} = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\left(R + \alpha R^2 + \beta R_{\mu\nu}R^{\mu\nu} + \ldots\right) \tag{10}$$

with coefficients determined by the entanglement spectrum.

## 5.3 Phenomenological Consequences

- **Holographic limits:** Mutual-information encoding implies an irreducible noise floor in precision interferometry.
- **Curvature corrections:** The higher-order terms in Eq. (10) may affect black hole quasinormal modes or early-universe cosmology.

- **Wave propagation:** Collective excitations exhibit modified dispersion: $\omega^2 = k^2[1+\epsilon(k)]$, with $\epsilon(k)$ suppressed by the microscopic scale.

### 5.4 Relation to Existing Approaches

Unlike Loop Quantum Gravity or Causal Set Theory, which quantize classical geometric properties (areas or partial orders), this framework keeps geometry entirely emergent from pure, non-spatial quantum information states. It shares conceptual ground with the ER=EPR conjecture [5], extending it to show that global spacetime geometry is an aggregate manifestation of sub-system entanglement profiles.

## 6 Conclusion

The Gravitational Baseline Principle provides a mathematically self-consistent conceptual path forward. By recognizing that gravity is the geometric framework of measurement rather than an active field, we bypass the contradictions of background-dependent quantization.

## References

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